# Exit Ticket · reflect and retrieve

## Exit Ticket

Calling a C# function that returns a new balance automatically changes the caller's original int variable.

## Use key vocabulary from memory

1. Explain refactoring in your own words.

2. Explain encapsulation in your own words.

3. Explain instance in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.



RHS COMPUTING / OCR H446 / LESSON 230 DRAFT · TEACHER REVIEW

# Do Now · retrieval

## Programming paradigms: applied comparison

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

How does OOP differ from a procedural program?

### Q2

What does a local variable mean?

### Q3

If a field is private-set, how can callers change it safely?

### Q4

Why does `Deposit(balance,5);` not update the caller's int balance if only a return value is produced?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

### Core idea 1

Procedural state — The caller passes an integer balance and explicitly stores the returned value.

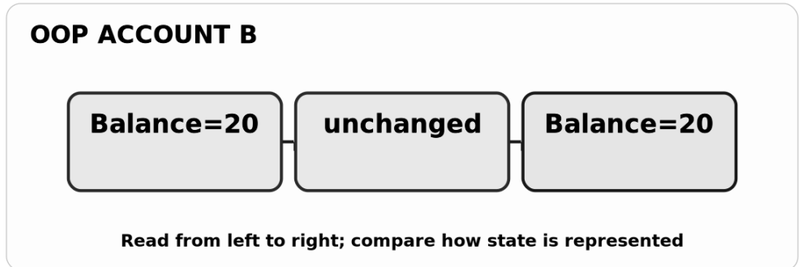
### Core idea 2

Object state — Account object A stores Balance internally and Deposit changes that object's state.

### Core idea 3

Separate instances — Account B stays at 20 when Account A changes to 15; the objects have independent state.

**Key visual · complete concept**




### Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

### Question 3

Compare one maintenance advantage and one risk of procedural and OOP implementations for a library system.

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Procedural AddFive(x) returns x+5. If caller x=2 discards the result, what is x?

## Question 2

Two objects have balances A=10, B=30. A.Deposit(7) and B.Deposit(2) run. Find both balances and describe why they differ.

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

refactoring — Restructuring code while preserving expected behaviour.

encapsulation — Controlled access to internal state.

instance — Individual object created from a class.

return value — Output returned to the caller.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

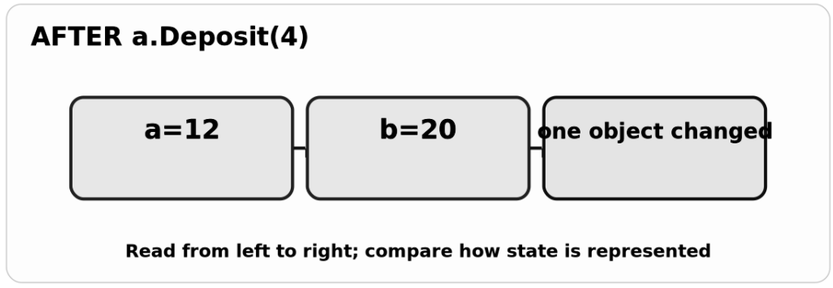
## Problem

Two accounts start at 8 and 20; only first account deposits 4. Trace both.

## Follow the worked method

1. Predict — Account a's Balance becomes 12.
2. Model response — Account b remains at 20 because its state is separate.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict
Deposit computes and returns 10, but its result is ignored.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.